

Why Big Data and Where did it come from?,

Why Big Data?

1. **Volume:** The sheer amount of data generated daily is immense. Businesses, social media, IoT devices, and other sources produce terabytes and petabytes of data. Traditional data processing tools can't handle such vast amounts.
2. **Variety:** Data comes in various formats – structured (databases), semi-structured (XML, JSON), and unstructured (text, images, videos). Big Data technologies are designed to handle and analyze these diverse types of data.
3. **Velocity:** Data is generated at unprecedented speeds. For example, social media updates, sensor data, and financial transactions produce streams of data that need real-time processing.
4. **Veracity:** Ensuring the quality and accuracy of data is crucial. Big Data tools help in cleaning, processing, and analyzing data to ensure it's reliable.
5. **Value:** Analyzing Big Data helps businesses and organizations derive actionable insights, improve decision-making, enhance customer experiences, and create competitive advantages.

Where Did Big Data Come From?

1. **Technological Advancements:** The development of advanced computing technologies, such as distributed computing and parallel processing, enabled the handling of large datasets.
2. **Digital Transformation:** The digitalization of various sectors (e.g., finance, healthcare, retail) led to massive amounts of data generation. The

proliferation of internet use, social media, and e-commerce also contributed significantly.

3. **Mobile Devices and IoT:** The rise of smartphones, wearables, and IoT devices generated continuous streams of data from various sources, adding to the Big Data ecosystem.
4. **Cloud Computing:** The availability of scalable cloud storage and computing resources made it feasible to store and process large datasets without substantial investments in on-premises infrastructure.
5. **Open Source Tools:** The development and adoption of open-source Big Data tools and frameworks (e.g., Hadoop, Spark) made it easier for organizations to implement Big Data solutions.
6. **Data-Driven Culture:** As organizations recognized the importance of data-driven decision-making, they invested in technologies and strategies to harness the potential of Big Data.

Evolution of Big Data

1. **Early Days:** The term "Big Data" began to gain traction in the early 2000s. However, the concept of managing and analyzing large datasets has been around since the advent of computers and databases.
2. **2000s:** The development of Hadoop by Doug Cutting and Mike Cafarella in 2006 was a significant milestone. Hadoop's ability to process large datasets using a distributed computing approach revolutionized Big Data processing.
3. **2010s:** The rise of NoSQL databases, real-time data processing frameworks, and machine learning algorithms further propelled the Big Data movement. Companies like Google, Facebook, and Amazon leveraged Big Data to optimize their operations and services.

4. **Present and Future:** Today, Big Data continues to evolve with advancements in AI, machine learning, and edge computing. The focus is on making Big Data more accessible, scalable, and efficient, enabling organizations to unlock deeper insights and drive innovation.

Big Data has become a crucial component in various industries, driving significant advancements and transforming how businesses and organizations operate.

Big Data Characteristics

Big Data contains a large amount of data that is not being processed by traditional data storage or the processing unit. It is used by many **multinational companies** to **process** the data and business of many **organizations**. The data flow would exceed **150 exabytes** per day before replication.

There are five v's of Big Data that explains the characteristics.

5 V's of Big Data

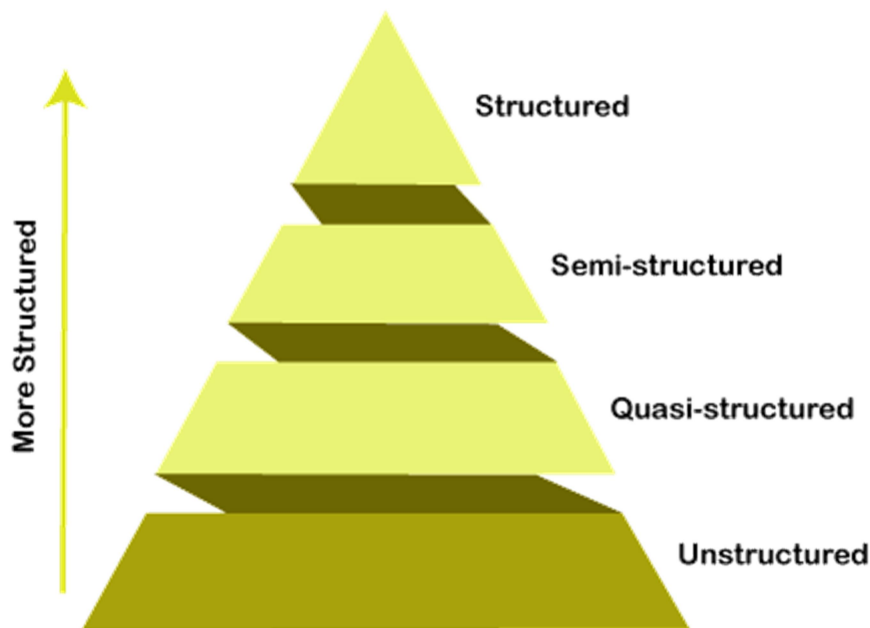
Volume

The name Big Data itself is related to an enormous size. Big Data is a vast 'volumes' of data generated from many sources daily, such as **business processes, machines, social media platforms, networks, human interactions**, and many more.

Facebook can generate approximately a **billion** messages, **4.5 billion** times that the "**Like**" button is recorded, and more than **350 million** new posts are uploaded each day. Big data technologies can handle large amounts of data.

Variety

Big Data can be **structured, unstructured, and semi-structured** that are being collected from different sources. Data will only be collected from **databases** and **sheets** in the past, But these days the data will comes in array forms, that are **PDFs, Emails, audios, SM posts, photos, videos, etc.**



The data is categorized as below:

- a. **Structured data:** In Structured schema, along with all the required columns. It is in a tabular form. Structured Data is stored in the relational database management system.
- b. **Semi-structured:** In Semi-structured, the schema is not appropriately defined, e.g., **JSON, XML, CSV, TSV**, and **email**. OLTP (**Online Transaction Processing**) systems are built to work with semi-structured data. It is stored in relations, i.e., **tables**.
- c. **Unstructured Data:** All the **unstructured files, log files, audio files**, and **image** files are included in the unstructured data. Some organizations

have much data available, but they did not know how to **derive** the value of data since the data is raw.

- d. **Quasi-structured Data:** The data format contains textual data with inconsistent data formats that are formatted with effort and time with some tools.

Example: Web server logs, i.e., the log file is created and maintained by some server that contains a list of **activities**.

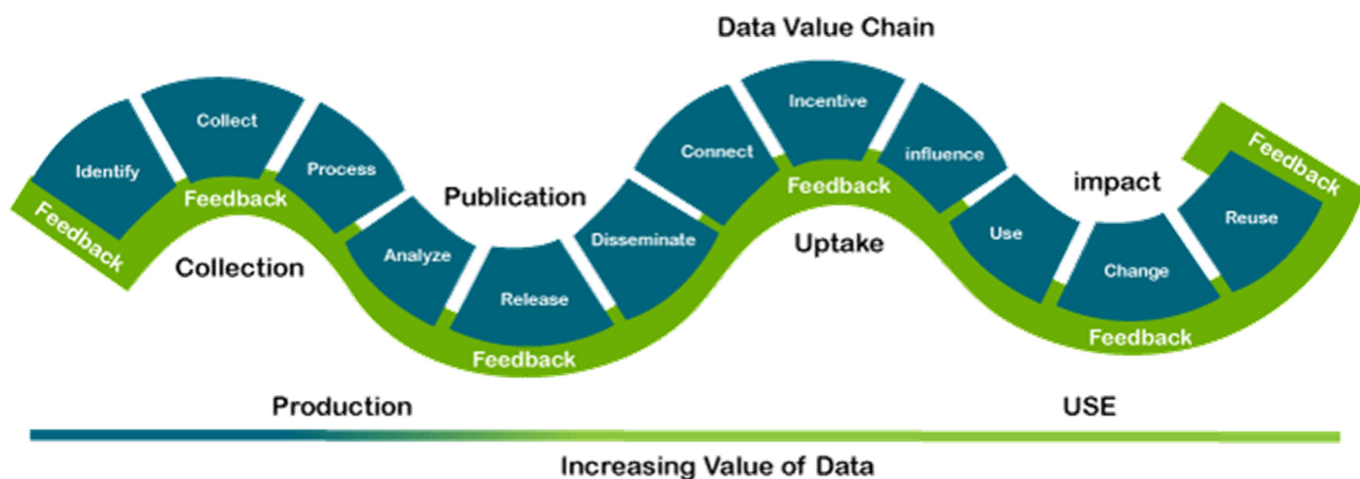
Veracity

Veracity means how much the data is reliable. It has many ways to filter or translate the data. Veracity is the process of being able to handle and manage data efficiently. Big Data is also essential in business development.

For example, **Facebook posts** with hashtags.

Value

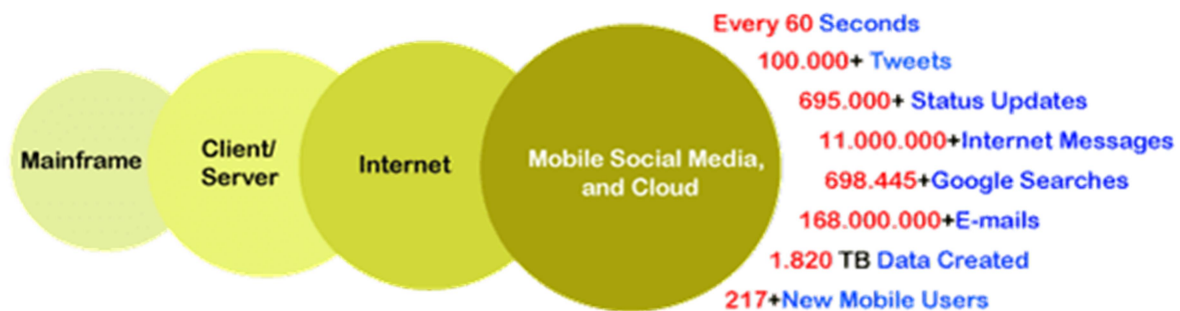
Value is an essential characteristic of big data. It is not the data that we process or store. It is **valuable** and **reliable** data that we **store, process,** and also **analyze**.



Velocity

Velocity plays an important role compared to others. Velocity creates the speed by which the data is created in **real-time**. It contains the linking of incoming **data sets speeds, rate of change, and activity bursts**. The primary aspect of Big Data is to provide demanding data rapidly.

Big data velocity deals with the speed at the data flows from sources like **application logs, business processes, networks, and social media sites, sensors, mobile devices, etc.**



Big Challenges with Big Data

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The challenges in **Big Data** are the real implementation hurdles. These require immediate attention and need to be handled because if not handled then the failure of the technology may take place which can also lead to some unpleasant result. Big data challenges include the storing, analyzing the extremely large and fast-growing data.

Some of the Big Data challenges are:

1. *Sharing and Accessing Data:*